

Module: Reverse Engineering

What is Reverse Engineering?

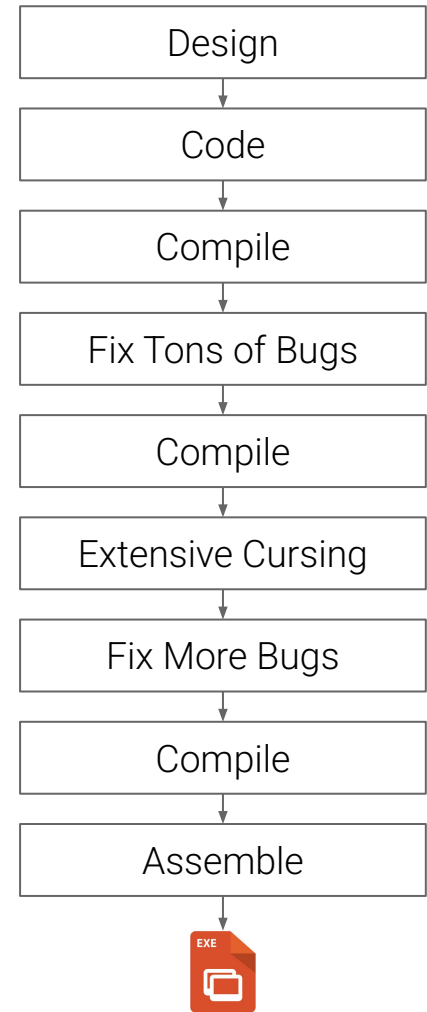
Yan Shoshitaishvili
Arizona State University

The Forward Engineering Process

"Forward Engineering" is an overloaded term, but in this context, it is the process of building a program.

1. Figure out what you want to code.
2. Code it.
3. Compile it (can include JIT).
4. Run it.

At every step, information is lost!



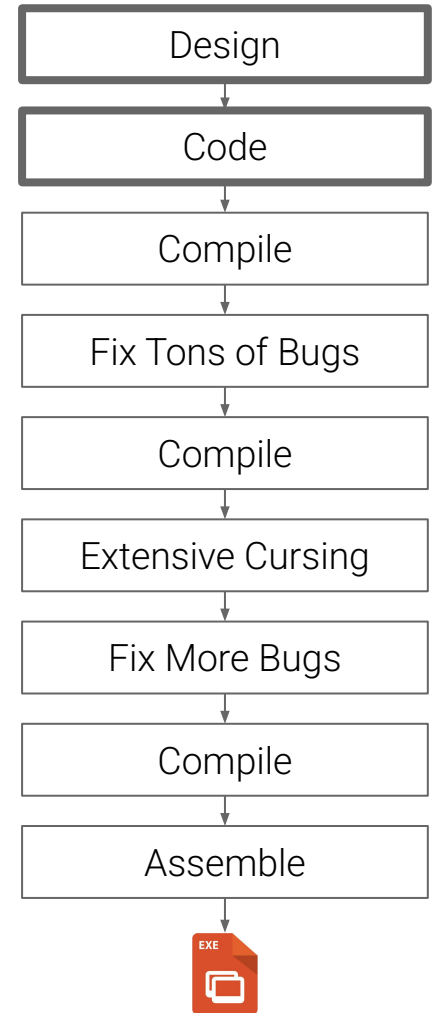
Discussion

What is lost in the transition between Design and Code?

Who wrote this bad code??

...

Oh, it was me...

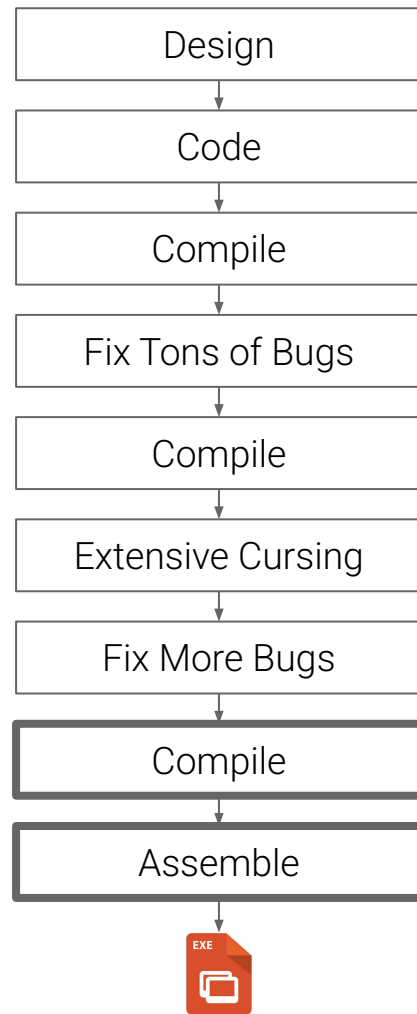


Discussion

What is lost in the compilation process?

- Comments
- Variable names.
- Function names.
- Structure (classes, structs, etc) data.
- Sometimes, entire algorithms (optimization)!

How do we get it back?

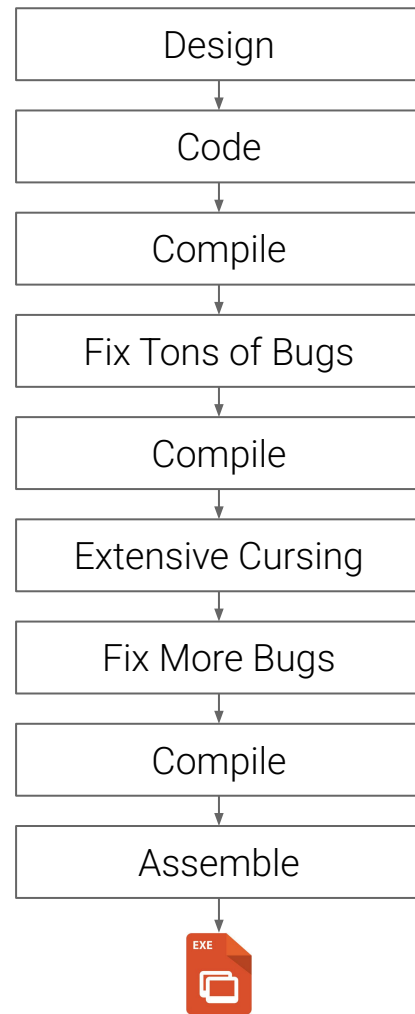


Forward Engineering Tools

Let's look at some tools:

- vim / an IDE
- gcc
- strings
- strip

Viola! An ELF is born.



The Reverse Engineering Process

Every step in the reverse-engineering process is imperfect and relies on some amount of human help.

Typically, a reverser use several reverse engineering tools to build up a mental model of the target software.

This art is the focus of this module: how do we reverse the design from the binary?

